

WAYNE COUNTY AP, KY, USA

WMO: 720379

Lat: 36.855N

Lon: 84.856W

Elev: 963

StdP: 14.19

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 63882

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 8.8	(c) 14.1	(d) -0.9	(e) 5.4	(f) 13.8	(g) 4.7	(h) 7.3	(i) 18.2	(j) 24.7	(k) 47.5	(l) 22.0	(m) 49.1	(n) 4.3	(o) 10	(p) 0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.6	(c) 92.8	(d) 75.2	(e) 90.6	(f) 75.1	(g) 88.6	(h) 74.6	(i) 79.1	(j) 87.5	(k) 77.7	(l) 86.2	(m) 76.5	(n) 84.9	(o) 6.1	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.9	145.1	84.5	75.4	137.7	82.5	74.0	131.6	80.7	43.3	87.3	41.7	86.4	40.4	84.7	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 21.0	(b) 18.7	(c) 16.7	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(4) 21.0	(5) 18.7	(6) 16.7	(7) -2.2	(8) 81.2	(9) 8.5	(10) 2.7	(11) -8.3	(12) 83.2	(13) -13.2	(14) 84.8	(15) -18.0	(16) 86.3	(17) -24.1	(18) 88.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.8	34.5	38.1	47.7	57.2	66.1	73.6	75.8	74.8	69.2	57.4	45.6	40.0		
	DBStd	17.00	12.59	11.46	10.93	9.18	7.69	4.80	4.93	4.76	6.42	9.01	9.54	9.75		
	HDD50	1641	502	359	178	36	1	0	0	0	0	32	194	339		
	HDD65	4254	945	753	541	261	86	4	1	1	32	269	585	776		
	CDD50	4111	22	27	105	253	502	707	800	769	576	260	62	28		
	CDD65	1248	0	0	4	28	121	261	336	305	158	32	3	0		
	CDH74	12259	0	1	60	365	1121	2528	3174	2944	1647	404	15	0		
	CDH80	4724	0	0	5	57	317	985	1366	1255	636	102	1	0		
(14) Wind	WSAvg	6.3	7.7	8.0	8.1	7.7	6.5	5.1	4.1	3.6	4.9	5.9	6.5	7.5		
(15) Precipitation	PrecAvg	51.8	5.1	3.9	5.0	4.3	5.0	4.7	4.7	4.1	3.3	2.9	3.5	5.4		
	PrecMax	63.3	8.1	7.6	10.8	9.2	9.4	9.5	7.5	8.0	6.3	5.2	6.9	14.3		
	PrecMin	40.0	2.4	1.5	2.1	1.2	2.0	2.0	2.2	1.3	0.3	0.7	0.7	2.4		
	PrecStd	6.6	1.3	1.6	2.2	1.8	1.9	2.0	1.6	1.8	1.9	1.3	1.4	2.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.1	71.5	80.2	84.3	89.5	93.3	98.4	96.9	94.6	88.5	76.3	68.6		
		MCWB	59.4	60.5	64.5	65.3	73.2	73.7	76.2	75.0	72.5	71.1	63.0	60.5		
	2%	DB	63.5	66.7	74.9	81.5	86.2	90.6	92.1	92.4	90.0	82.5	70.3	64.7		
		MCWB	57.1	57.6	61.4	64.7	71.1	74.6	76.6	75.6	72.8	67.5	59.5	58.3		
	5%	DB	59.9	62.3	70.3	78.6	83.7	88.4	89.9	89.9	87.2	78.7	66.3	60.5		
		MCWB	54.6	55.0	58.5	63.0	70.1	74.2	76.4	75.1	72.1	65.0	57.3	55.0		
	10%	DB	55.2	58.0	66.3	75.0	81.0	86.0	87.6	87.1	83.5	74.2	63.0	57.3		
		MCWB	50.5	51.4	55.8	62.1	69.1	73.3	75.6	74.5	69.9	63.6	55.3	52.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.2	62.9	65.7	70.2	75.8	78.7	81.5	81.1	76.9	74.1	66.2	63.1		
		MCDB	65.5	68.8	76.4	81.3	85.9	87.9	89.3	90.6	86.3	83.9	72.7	67.7		
	2%	WB	58.4	60.1	63.3	67.9	73.3	77.3	79.7	78.4	75.3	70.9	63.3	59.3		
		MCDB	62.4	65.7	73.0	77.6	82.7	86.5	87.9	87.1	85.9	78.1	67.4	62.8		
	5%	WB	55.3	56.5	60.8	65.4	71.9	75.8	78.3	77.1	73.7	67.9	59.8	56.4		
		MCDB	59.4	60.5	68.1	74.0	81.2	85.2	86.3	85.8	83.4	75.0	63.9	59.6		
	10%	WB	51.1	52.6	58.1	63.4	70.4	74.6	77.0	75.7	72.0	65.3	56.3	52.9		
		MCDB	54.3	56.8	64.5	71.7	78.7	83.4	84.7	83.9	80.1	72.1	61.6	56.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.6	20.9	23.0	25.7	22.0	22.6	21.6	23.0	23.7	25.4	24.4	20.1		
		MCDBR	22.2	26.4	28.5	31.0	25.2	25.3	25.4	26.3	28.6	30.3	28.5	23.9		
	5% WB	MCWBR	17.1	18.7	17.1	16.1	11.9	11.2	11.2	11.4	12.7	15.9	18.1	18.6		
		MCWBR	19.3	21.5	24.5	24.1	22.4	21.5	20.4	21.7	23.6	22.9	19.9	21.3		
(40) Clear-Sky Solar Irradiance	taub		0.308	0.318	0.351	0.394	0.445	0.465	0.499	0.503	0.413	0.362	0.324	0.305		
	taud		2.499	2.460	2.411	2.284	2.176	2.152	2.081	2.056	2.289	2.425	2.497	2.517		
	Ebn at Noon		281	292	290	281	268	261	251	247	266	271	272	275		
	Edh at Noon		26	30	35	42	47	48	51	51	39	31	25	24		
(44) All-Sky Solar Radiation	RadAvg		657	850	1207	1594	1772	1950	1833	1727	1444	1117	796	571		
	RadStd		69	116	113	117	155	112	99	90	149	142	86	56		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (21 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+243	+157

Nomenclature: See separate page